SS1 PHYSICS LESSON NOTES (THIRD TERM)
WEEK 2: GAS LAWS

. DEFINITION OF A GAS

A gas is a state of matter that has **no fixed shape or volume**.
It completely fills the container it is placed in.

Key Properties of Gases

- No fixed shape
- No fixed volume
- Highly compressible
- Particles move randomly and rapidly
- Very weak intermolecular forces

2. ASSUMPTIONS OF KINETIC THEORY OF GASES

1. Gases are made up of a large number of tiny particles (atoms or molecules) in constant random motion
2. The volume of gas particles is negligible compared to the container volume
3. There are no attractive or repulsive forces between gas particles
4. Collisions between particles and container walls are perfectly elastic
5. Average kinetic energy is directly proportional to absolute temperature

Key Formula (Kinetic Energy):

KE(avg) = (3/2) kT

Where:

- $k = 1.38 \times 10^{-23}$ J/K (Boltzmann constant)
- T = temperature in Kelvin

3. BOYLE'S LAW (PRESSURE–VOLUME LAW)

At constant temperature, the volume of a fixed mass of gas is **inversely proportional to its pressure**.

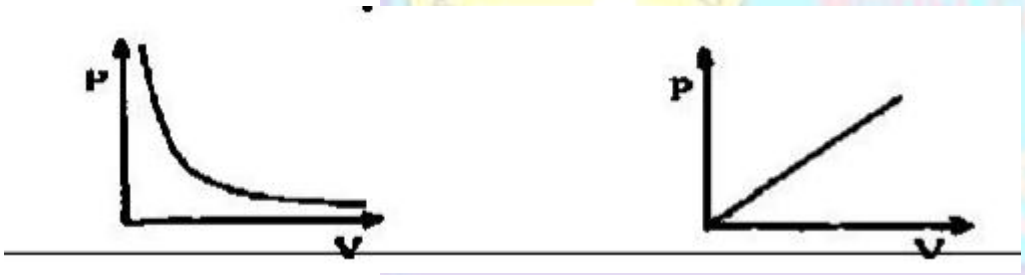
Formula:

$P \propto 1/V$
$PV = \text{constant}$

Equation:

$P_1V_1 = P_2V_2$

Graphical Representation



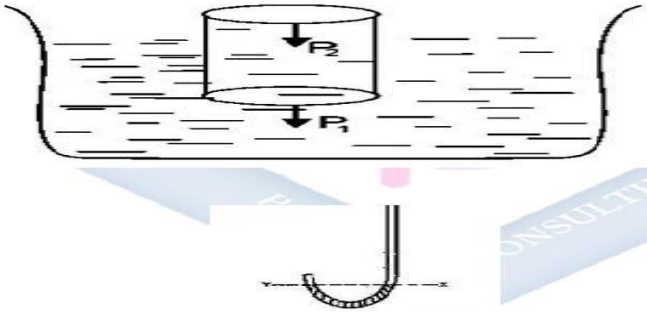
- P vs V → Hyperbola
- P vs 1/V → Straight line

Experimental Setup

A syringe is sealed and compressed. As volume decreases, pressure increases.

Experimental Demonstration:

Use a syringe sealed at the nozzle. Push the plunger while observing the pressure change.



Worked Example

In a J-tube, 30 cm³ of air is trapped when atmospheric pressure is 75 cmHg.
When 15 cm Hg is added above X, find new volume.

Solution:

Using Boyle's Law:

$P_1V_1 = P_2V_2$

$P_1 = 75$ cmHg
$V_1 = 30$ cm³
$P_2 = 75 + 15 = 90$ cmHg
$V_2 = ?$

$30 \times 75 = 90 \times V_2$

$V_2 = (30 \times 75) / 90$
$V_2 = 25$ cm³

4. CHARLES' LAW (VOLUME–TEMPERATURE LAW)

At constant pressure, volume is directly proportional to absolute temperature.

Formula:

$V \propto T$

$V/T = \text{constant}$

Equation:

$V_1/T_1 = V_2/T_2$

⚠ Temperature must be in Kelvin:
$K = °C + 273$

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5. PRESSURE LAW (GAY-LUSSAC'S LAW)

At constant volume, pressure is directly proportional to absolute temperature.

Formula:

$P \propto T$

$P/T = \text{constant}$

Equation:

$P_1/T_1 = P_2/T_2$

Worked Example

A gas at 27°C has its pressure doubled at constant volume. Find final temperature.

Solution:

$T_1 = 27 + 273 = 300$ K

$P_2 = 2P_1$

Using:

$P_1/T_1 = P_2/T_2$

$P_1/300 = 2P_1/T_2$

Cancel P_1:

$1/300 = 2/T_2$

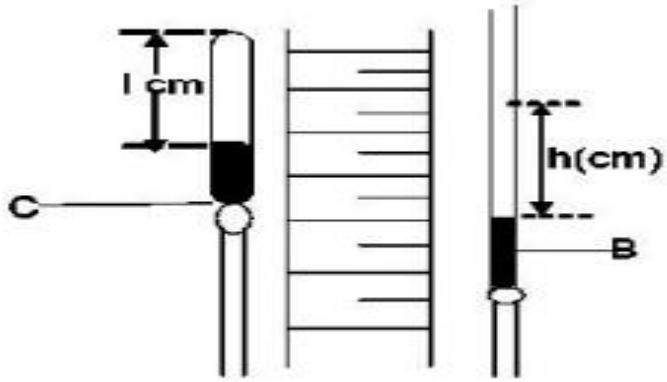
$T_2 = 600$ K

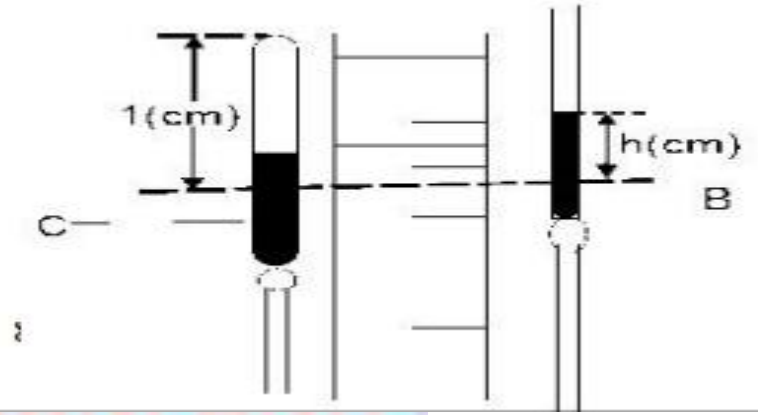
Convert to °C:

$T_2 = 600 - 273 = 327$°C

6. GENERAL GAS LAW

Combining Boyle's, Charles's and Pressure Laws: This is the relationship between the volume and pressure of a given mass of gas at constant temperature.





Atmospheric pressure = H cm Hg P = (H - h) cm Hg Atmospheric pressure = H cm Hg P = (H + h) cm Hg Some dry air is enclosed in a glass tube of uniform cross-section attached by rubber tubing to a mercury reservoir. By raising or lowering the open tube (B), the length of the air column in the closed tube (C) can be varied. The length read on the metre scale is measured as the volume of the gas (since tube is of uniform cross-section). The difference (h) in the levels of mercury at C and B is read for each value in length of gas. The atmospheric pressure (H) is read from a fort in barometer. When B is lower than C, pressure is, P = (H - h) cm Hg When C is higher than B, pressure is P = (H + h) cm Hg **various values of l** and h is obtained by lowering and raising tube B. A graph of P is plotted against V(L cm). It gives a result that shows it varies inversely. PVT = constant

Formula:

$(P_1V_1) / T_1 = (P_2V_2) / T_2$

Used when pressure, volume, and temperature all change.

Worked Example

$V_1 = 76\ cm^3$
$P_1 = 80$ cmHg

$T_1 = 300$ K
$P_2 = 760$ mmHg
$T_2 = 273$ K
$V_2 = ?$

Solution:

$(P_1V_1)/T_1 = (P_2V_2)/T_2$

Substitute:

$(80 \times 76)/300 = (760 \times V_2)/273$

Cross multiply:

$V_2 = (80 \times 76 \times 273) / (300 \times 760)$

$V_2 = 72.8$ cm³

7. IDEAL GAS EQUATION

Formula:

$PV = nRT$

Where:

- P = pressure (Pa)
- V = volume (m³)
- n = number of moles
- R = 8.314 J/mol·K
- T = temperature (K)

8. MOLAR VOLUME & AVOGADRO'S LAW

Avogadro's Law

Equal volumes of gases at the same temperature and pressure contain equal numbers of molecules.

Molar Volume

At STP:

- 1 mole of gas = 22.4 L = 22.4 dm^3

Worked Example

Find pressure of 1 mole of gas:

$V = 10^{-2}\ m^3$
$T = 27°C = 300\ K$
$n = 1$
$R = 8.3$

$PV = nRT$

$P \times 10^{-2} = 1 \times 8.3 \times 300$

$P = (8.3 \times 300) / 10^{-2}$

$P = 2.49 \times 10^5\ Pa$

9. REAL VS IDEAL GASES

Ideal Gas

- No intermolecular forces

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- Perfectly elastic collisions
- Negligible particle volume

Real Gas

- Deviates at high pressure and low temperature
- Molecules attract each other
- Explained by Van der Waals equation

10. APPLICATIONS OF GAS LAWS

- Airbags (Boyle's law)
- Hot air balloons (Charles' law)
- Pressure cookers (Pressure law)
- Weather balloons
- Human breathing system

EVALUATION

1. State 3 assumptions of kinetic theory of gases
2. Derive general gas law
3. Solve: P_1=2 atm, V_1=3 L, T_1=300 K, P_2=4 atm, T_2=600 K, find V_2
4. Why do gases deviate from ideal behavior?
5. Define molar volume

ASSIGNMENT

1. Summarize Van der Waals equation
2. 5.6 g helium → number of moles (He = 4 g/mol)
3. Volume 44.8 L at STP → moles
4. 0.5 mol gas, P = 2 atm, T = 300 K → find volume
5. Gas law combined variation problem (full calculation given in p

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